MA∗508 Time Series

Exam-like exercises for L4

Several exercises are given below. Solutions of exercises 1 and 4 are presented for demonstration purposes. Please, solve the remaining exercises in a similar manner. You may use various convergence tests `https://en.wikipedia.org/wiki/Convergence_tests`

1. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = e^{-\alpha|t|}$ to be short-range or long range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty}(e^{-\alpha})^k = \sum_{k=1}^{\infty} q^k$ where $q = e^{-\alpha}$. The last sum is a geometric series, therefore $1 + \sum_{k=1}^{\infty} q^k = \frac{1}{1-q}$. Thus, the sum is finite, which means that $R(t) = e^{-\alpha|t|}$ is short-range dependent.

2. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = e^{-\alpha t^2}$ to be short-range or long range dependent.

3. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = e^{-\alpha|t|^{1.6}}$ to be short-range or long range dependent.

4. Determine whether the covariance function $R(t) = 1/(1+|t|)^3$ is short-range or long range dependent.

   The *limit comparison test* says that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} \frac{1}{(1+k)^3}$ converges/diverges if and only if $\sum_{k=1}^{\infty} \frac{1}{k^3}$ converges/diverges because the ratio of the k-th summand of the first sum to the k-th summand of the second sum tends to 1 as $k \to \infty$, i.e. $\lim_{k\to\infty} \frac{1}{(1+k)^3} \Big/ [\frac{1}{k^3}] = \lim_{k\to\infty} \frac{k^3}{(1+k)^3} = \lim_{k\to\infty} \frac{1}{(1/k+1)^3} = 1$. From the theory on convergence of series, we know $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ converges if $\alpha > 1$ and diverges otherwise. Thus, the covariance function $R(t) = 1/(1+|t|)^3$ is short-range dependent since $\alpha = 3 > 1$.

5. Determine whether the covariance function $R(t) = 1/(1+t^2)$ is short-range or long-range dependent.

6. Determine whether the covariance function $R(t) = 1/(1+\sqrt{|t|})^2$ is short-range or long-range dependent.

7. Determine whether the covariance function $R(t) = 4/(5+5\sqrt{|t|})$ is short-range or long-range dependent.